Light as an Electromagnetic Wave

Part II: Light Behavior and Properties

The following subjects will be covered

- Reflection and Transmission of Light.
- Light Polarization.
- Light-matter Interaction.
- Scattering of Light.

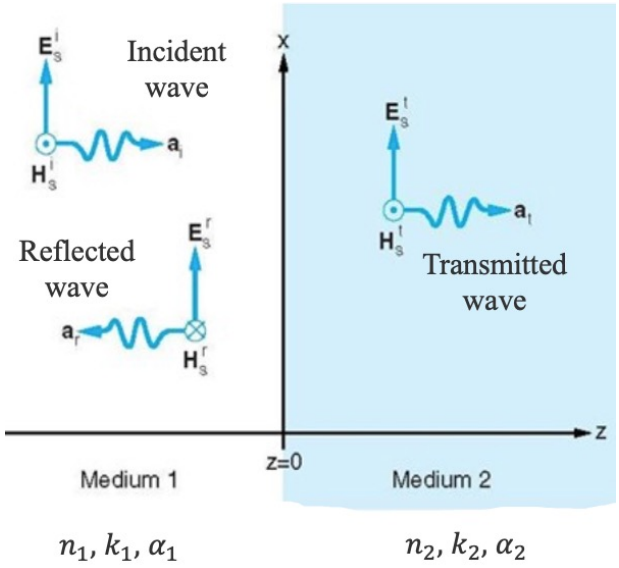
General Media

Electric Fields

$$\overrightarrow{E_i}(z) = E_o\, e^{-\alpha_1 z}\, e^{-jk_1 z}\, \vec{a}_x$$

$$\overrightarrow{E_r}(z) = \Gamma\, E_o\, e^{+\alpha_1 z}\, e^{+jk_1 z}\, \vec{a}_x$$

$$\overrightarrow{E_t}(z) = \tau E_o\, e^{-\alpha_2 z}\, e^{-jk_2 z}\, \vec{a}_x$$



Reflection coefficient

$$\Gamma = \frac{n_1 - n_2}{n_1 + n_2}$$

Transmission coefficient

$$\tau = \frac{2n_1}{n_1 + n_2}$$

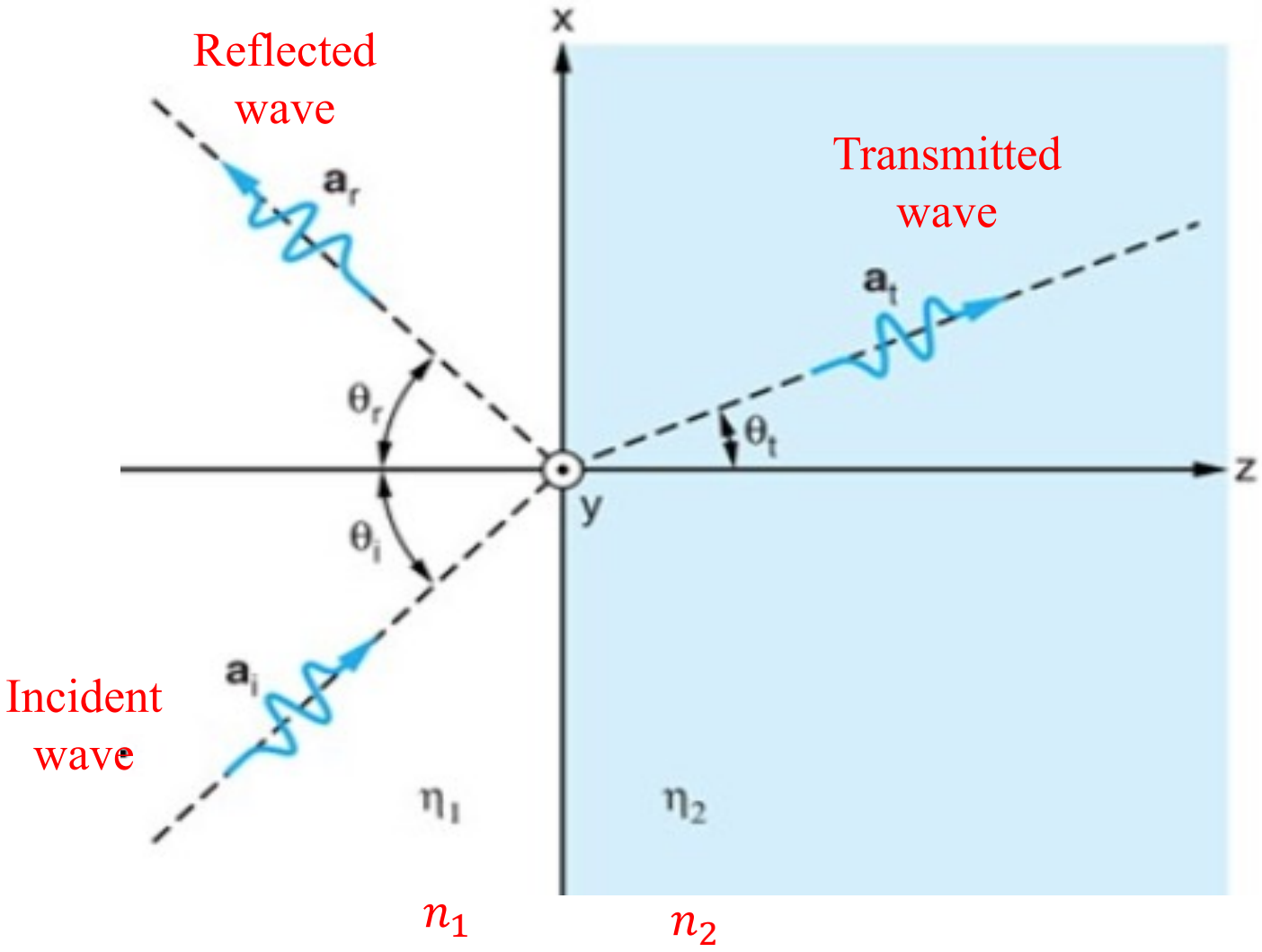
(n_1 and n_2 complex quantities)

$$\vec{S}^i_{ave} = \frac{|E_o|^2}{2}\, e^{-2\alpha_1 z}\, Re\left(\frac{1}{\eta_1}\right)\, \vec{a}_z$$

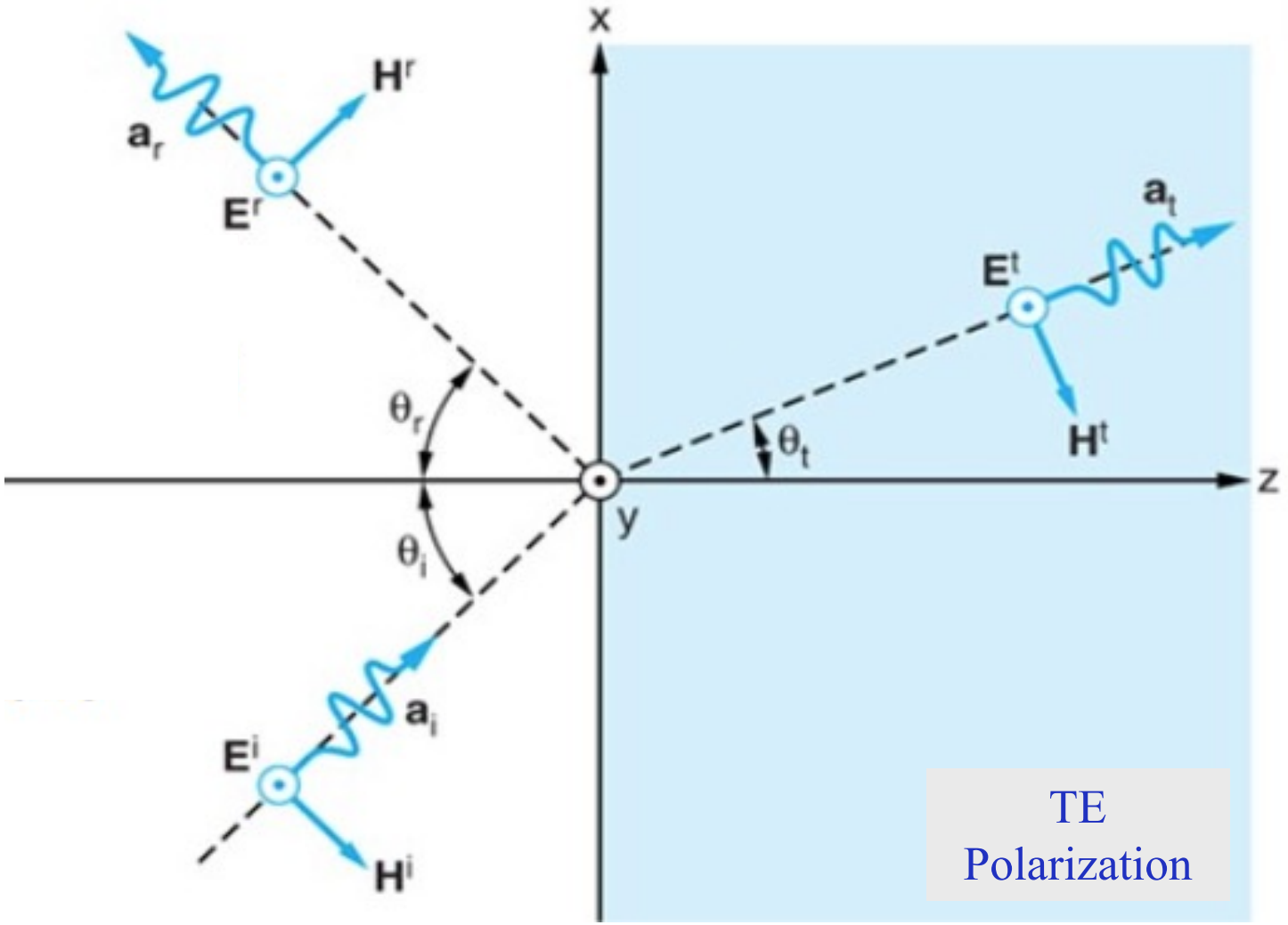
$$\vec{S}^r_{ave} = |\Gamma|^2\, \frac{|E_o|^2}{2}\, e^{+2\alpha_1 z}\, Re\left(\frac{1}{\eta_1}\right)\, (-\vec{a}_z)$$

$$\vec{S}^t_{ave} = |\tau|^2\, \frac{|E_o|^2}{2}\, e^{-2\alpha_2 z}\, Re\left(\frac{\eta_1}{\eta_2}\right)\, \vec{a}_z$$

Light Reflection and Transmission

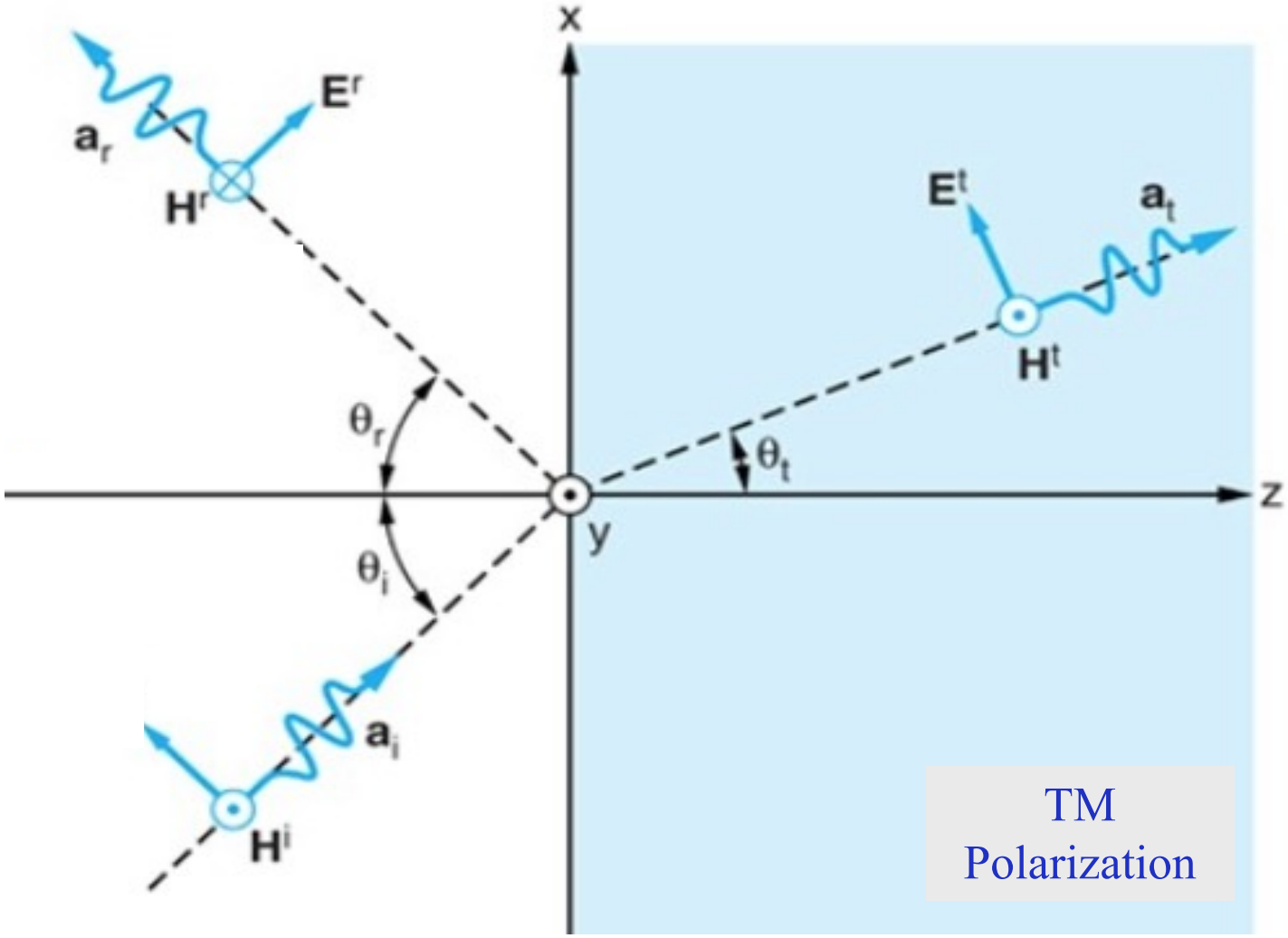


Single Interface: Oblique Incidence

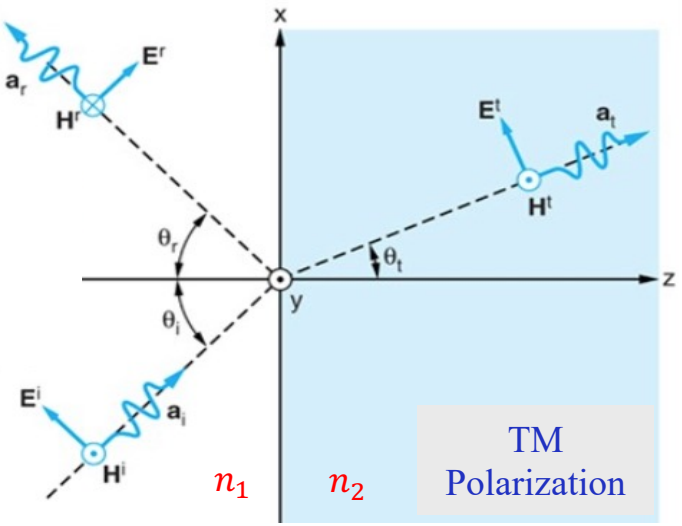


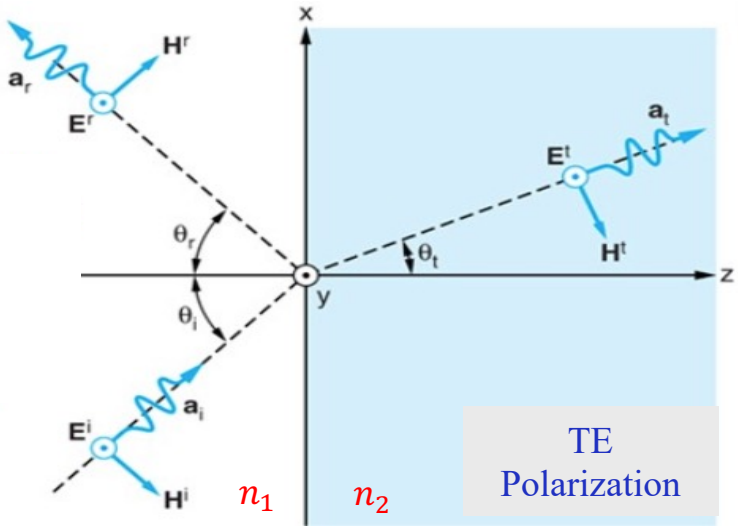
n_1 n_2

Single Interface: Oblique Incidence



Single Interface: Oblique Incidence





$$\Gamma_{TM} = \frac{n_1\,cos\theta_t - n_2\,cos\theta_i}{n_1\,cos\theta_t + n_2\,cos\theta_i}$$

$$\Gamma_{TM} = \frac{\tan(\theta_t - \theta_i)}{\tan(\theta_t + \theta_i)}$$

$$\tau_{TM} = \frac{2\,n_1\,cos\theta_i}{n_1\,cos\theta_t + n_2\,cos\theta_i}$$

$$\tau_{TM} = (1 + \Gamma_{TM})\frac{cos\theta_i}{cos\theta_t}$$

Snell's Law

$$sin\theta_t = \frac{n_1}{n_2}\,sin\theta_i$$

$$R = |\Gamma|^2$$

$$T = 1 - R$$

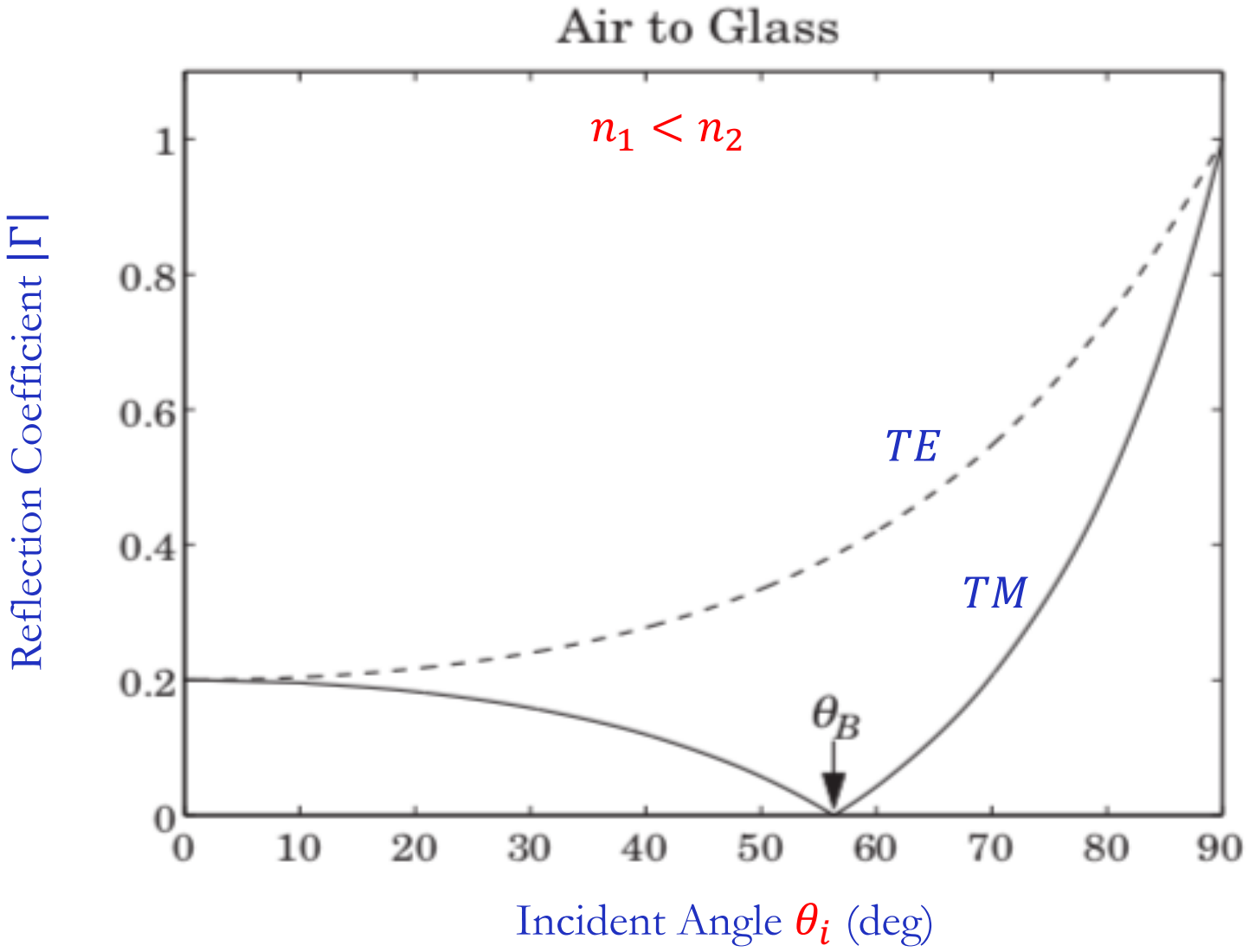
$$\Gamma_{TE} = \frac{n_1\,cos\theta_i - n_2\,cos\theta_t}{n_1\,cos\theta_i + n_2\,cos\theta_t}$$

$$\Gamma_{TE} = \frac{\sin(\theta_t - \theta_i)}{\sin(\theta_t + \theta_i)}$$

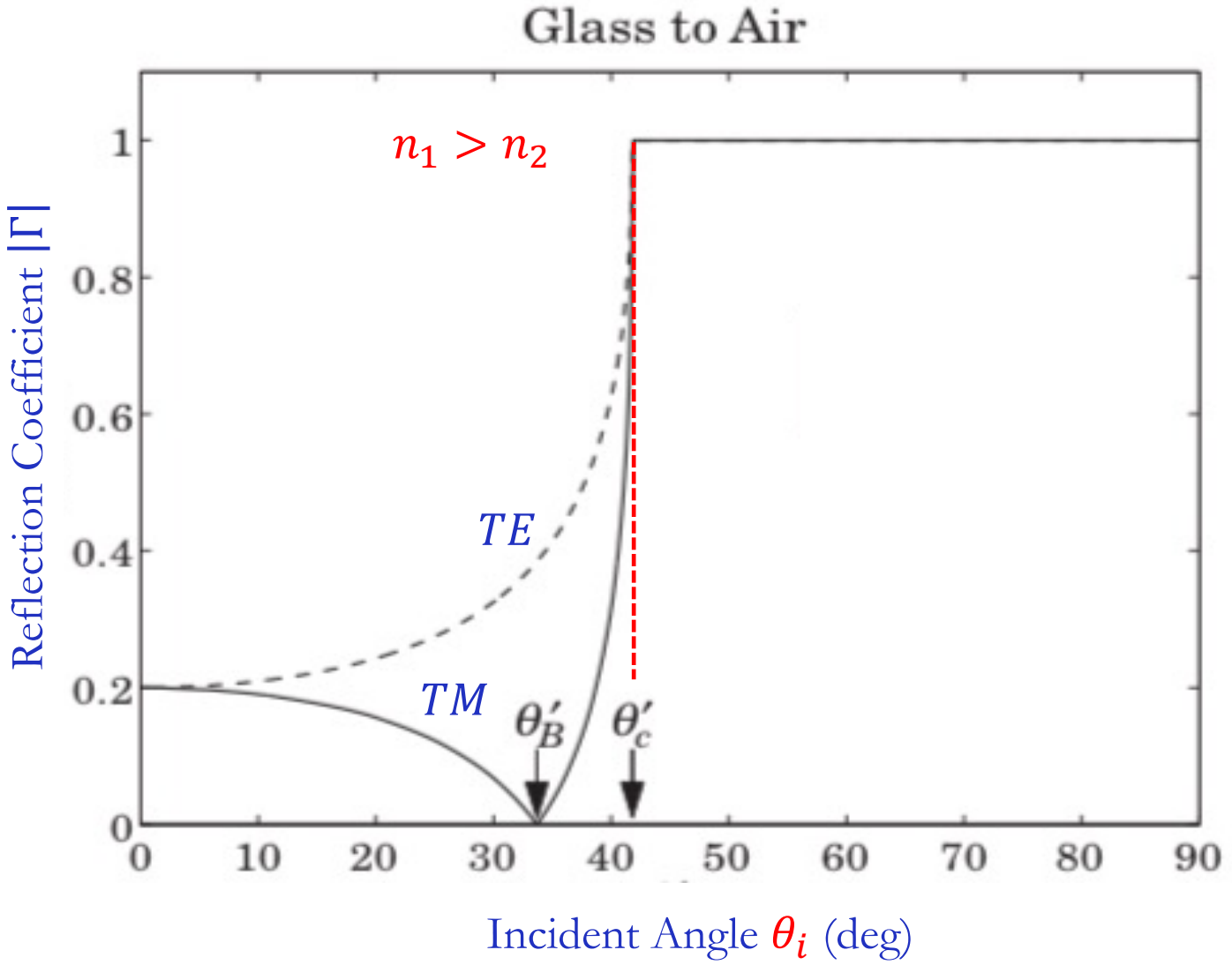
$$\tau_{TE} = \frac{2\,n_1\,cos\theta_i}{n_1\,cos\theta_i + n_2\,cos\theta_t}$$

$$\tau_{TE} = 1 + \Gamma_{TE}$$

Single Interface: Oblique Incidence



Single Interface: Oblique Incidence



Single Interface: Oblique Incidence

1) Total Transmission. $|\Gamma| = 0$, Only TM polarization.

Incident Angle $\theta_i = \tan^{-1}(n_2/n_1)$ Brewster Angle θ_B

In this case $\theta_i + \theta_t = 90^o$

Air-Glass: $\theta_B = \tan^{-1}(1.52/1)$

$\theta_B = 56.7^o$

2) Total Reflection $|\Gamma| = 1$, Both TE and TM polarizations, for $n_1 > n_2$ only.

Incident Angle $\theta_i = \sin^{-1}(n_2/n_1)$ Critical Angle θ_c

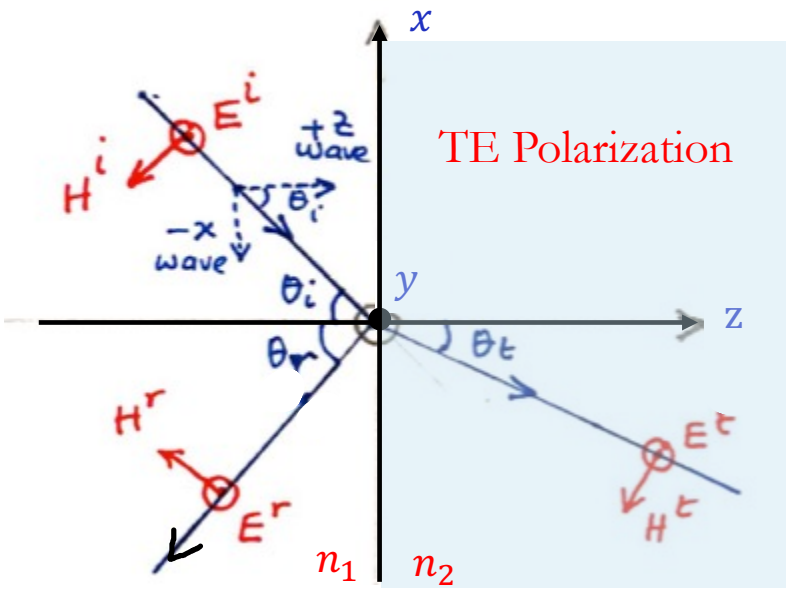
Happens for any angle $\theta_i \geq \theta_c$

Glass-Air: $\theta_c = \sin^{-1}(1/1.52)$

$\theta_c = 41.1^o$

Examples

A single-frequency plane wave is incident obliquely from air onto a planar interface with a non-magnetic medium having a refractive index of $\sqrt{2}$. The angle of incidence is 54.7°. Find the reflection and the transmission coefficients. Also find reflectance and transmittance.



$$\theta_i = 54.7°$$

From Snell's law, $n_1 \sin\theta_i = n_2 \sin\theta_t$

$$\sin\theta_t = \frac{n_2}{n_1}\sin\theta_i$$

$$\theta_t = 35.3°$$

$$\Gamma_{TE} = \frac{\sin(\theta_t - \theta_i)}{\sin(\theta_t + \theta_i)} = -0.333$$

$$R = (-0.333)^2 = 0.111$$

$$\tau_{TE} = 1 + \Gamma_{TE} = 0.667$$

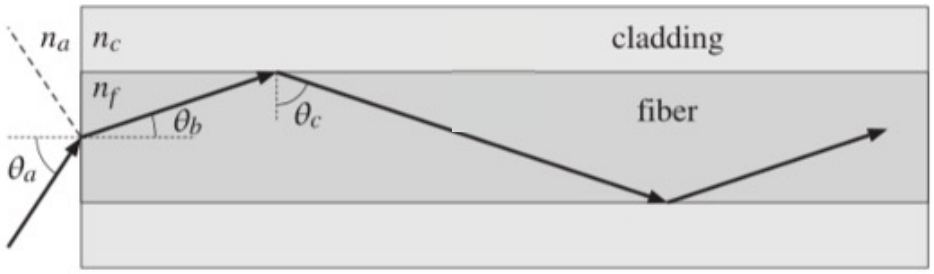
$$T = 1 - R = 0.889$$

$$\vec{E} = \left[\frac{1}{\sqrt{2}}\vec{a}_y\right] e^{-j\beta(z-\sqrt{2}x)/\sqrt{3}} \quad V/m$$

Examples

The Optical Fiber.

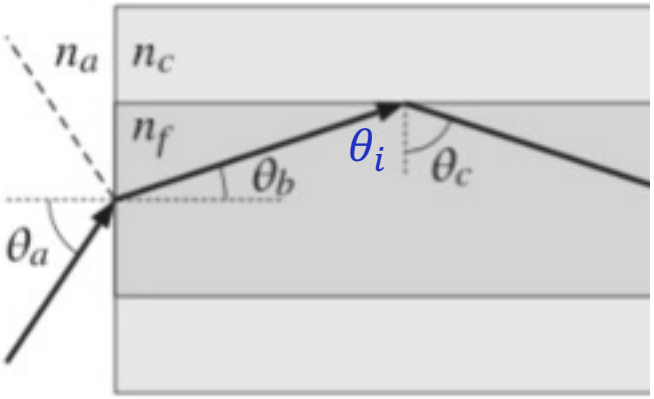
The acceptance angle θ_a should correspond to the critical angle inside the fiber at the cladding-fiber interface. $\theta_i = \theta_c = 90 - \theta_b$



Snell's Law $n_a \sin\theta_a = n_f \sin\theta_b$

$$\sin\theta_a = \frac{n_f}{n_a}\sin\theta_b = \frac{n_f}{n_a}\sin(\theta_c + 90)$$

$$\sin\theta_a = \frac{n_f}{n_a}\cos\theta_c = \frac{n_f}{n_a}\sqrt{1-(\sin\theta_c)^2}$$



Critical angle condition

$$\sin\theta_c = \frac{n_c}{n_f} \Rightarrow \sin\theta_a = \frac{n_f - n_c}{n_a}$$

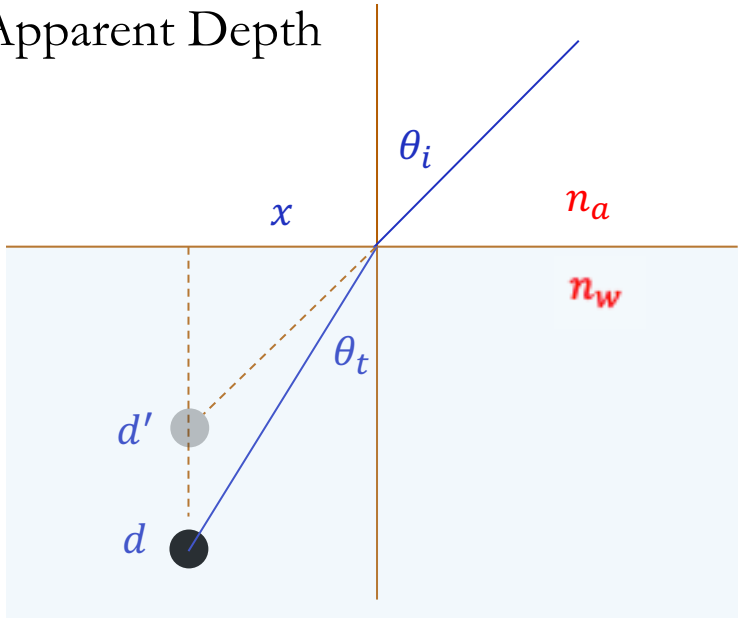
$n_a = 1, n_f = 1.49, n_c = 1.48$

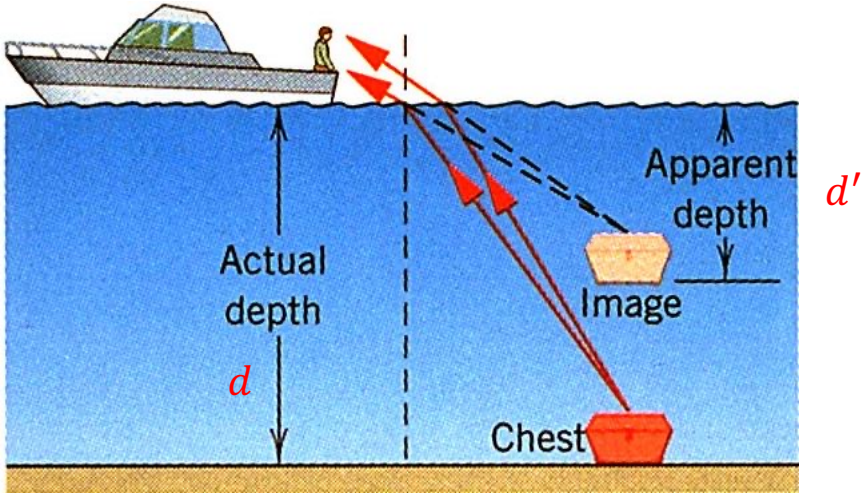
$\theta_c = 83.4°$

$\theta_a = 9.9°$

Examples

Apparent Depth





$$d = x \cot\theta_t$$

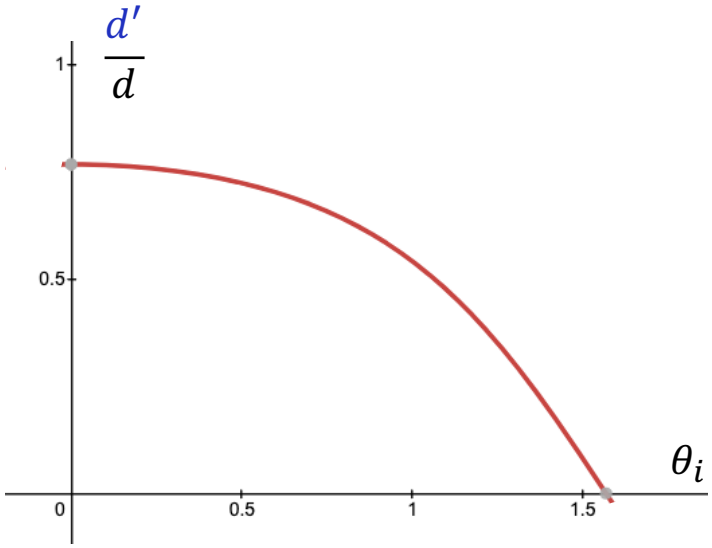
$$d' = x \cot\theta_i$$

$$n_a \sin\theta_i = n_w \sin\theta_t$$

$$\frac{d'}{d} = \frac{\cot\theta_i}{\cot\theta_t} = \frac{\cos\theta_i}{\cos\theta_t} \cdot \frac{\sin\theta_t}{\sin\theta_i}$$

$$\frac{d'}{d} = \frac{n_a}{n_w} \cdot \frac{\cos\theta_i}{\cos\theta_t}$$

$$\frac{d'}{d} = \frac{\cos\theta_i}{\sqrt{n_w^2 - (\sin\theta_i)^2}}$$



Anti-reflection Coating

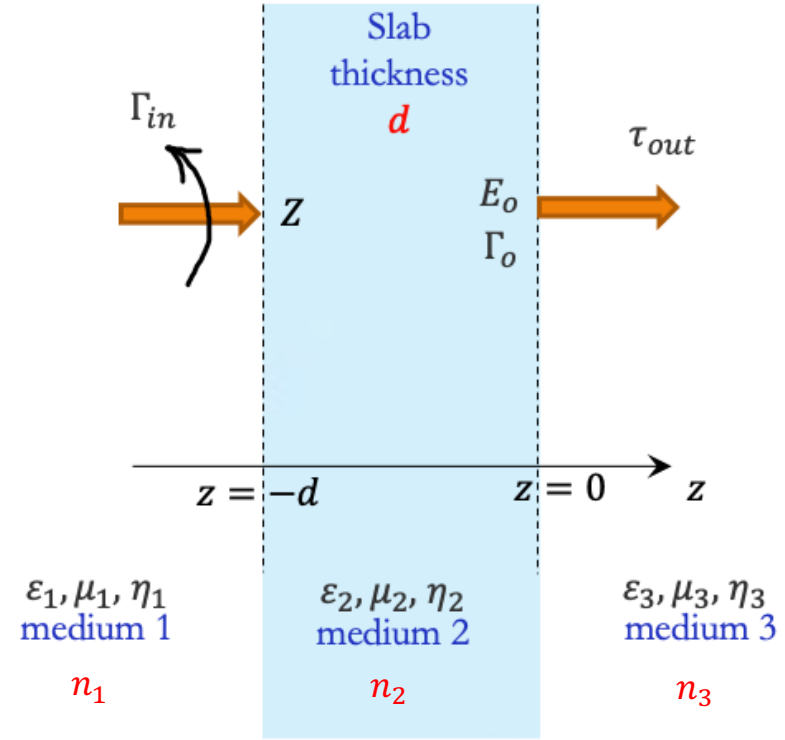
Let's consider normal incidence only. All media are lossless.

$$\Gamma_{in} = \frac{\Gamma_{12} + \Gamma_{23}\, e^{-j2k_2 d}}{1 + \Gamma_{12}\Gamma_{23}\, e^{-j2k_2 d}}$$

$$\Gamma_{12} = \frac{n_1 - n_2}{n_1 + n_2} \qquad \Gamma_{23} = \frac{n_2 - n_3}{n_2 + n_3}$$

For Γ_{12} and Γ_{23} samll

$$\Gamma_{in} \approx \Gamma_{12} + \Gamma_{23}\, e^{-j2k_2 d}$$



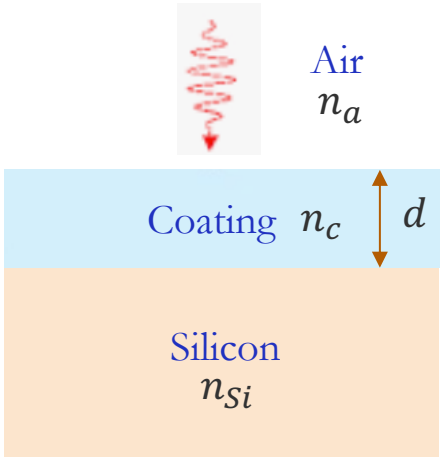
Similarly, $$\tau_{out} = \frac{\tau_{12}\tau_{23}\, e^{-jk_2 d}}{1 + \Gamma_{12}\Gamma_{23}\, e^{-j2k_2 d}}$$

$$\tau_{12} = 1 + \Gamma_{12}$$

$$\tau_{23} = 1 + \Gamma_{23}$$

Examples

1) A coating is needed on a silicon solar cell ($n_{Si} = 4.4 - 3.8$) such that incident light at mid-visible wavelengths (550 nm) from air is not reflected. Find the required coating thickness and material.



$\Gamma_{in} = 0$ ➡ $\left|\Gamma_{a-c} + \Gamma_{c-Si}\, e^{-j2k_c d}\right| = 0$

$$\Gamma_{a-c} = \frac{n_a - n_c}{n_a + n_c} \qquad \Gamma_{c-Si} = \frac{n_c - n_{Si}}{n_c + n_{Si}}$$

Look at $2k_c d = 2\frac{2\pi}{\lambda_c} d = \pi \frac{4}{\lambda_c} d$

If we take $d = \frac{\lambda_c}{4}$ Quarter-wavelength Transformer then $2k_c d = \pi$

➡ $e^{-j2k_c d} = e^{-j\pi} = -1$ ➡ $\left|\frac{n_a - n_c}{n_a + n_c} - \frac{n_c - n_{Si}}{n_c + n_{Si}}\right| = 0$

$$\left|\frac{(n_a - n_c)(n_c + n_{Si}) - (n_c - n_{Si})(n_a + n_c)}{(n_a + n_c)(n_c + n_{Si})}\right| = \left|\frac{2(n_a n_{Si} - n_c^2)}{(n_a + n_c)(n_c + n_{Si})}\right| = 0 \;\Rightarrow\; 2|(n_a n_{Si} - n_c^2)| = 0$$

If we take $n_c = \sqrt{n_a n_{Si}}$ ➡ $\Gamma_{in} = 0$

Coating material $n_c \approx 2$

$d = \frac{\lambda_c}{4} \approx 140\ nm$ Anti-reflection Coating thickness

Examples

2) Design a ¼-wavelength antireflection coating for a glass substrate ($n_g = 1.5$) . The coating is supposed to work for mid-visible range ($\lambda \approx 550\ nm$) .

Using the quarter wavelength condition $n_c = \sqrt{n_a n_g} = \sqrt{1.5} = 1.22$

$$d = \frac{\lambda_c}{4} = 112.7\ nm$$



The solution is not *that* easy!

Practically, the closest materials to the requirement are Cryolite ($n = 1.35$) and Magnesium Fluoride ($n = 1.38$, more durable for this task and more abundant).

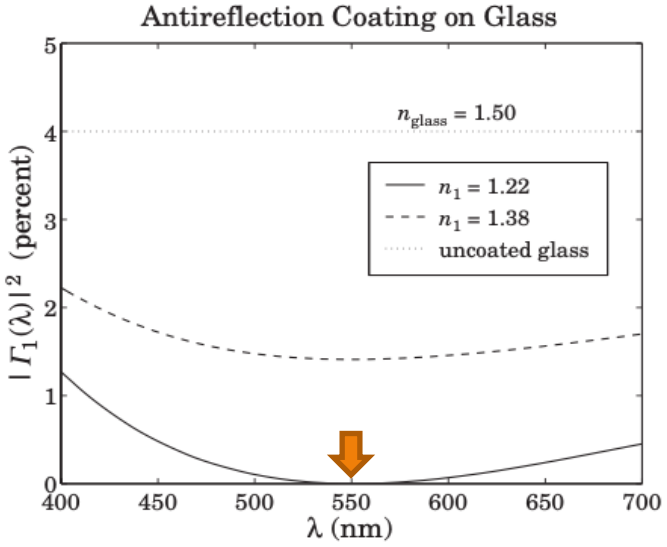
Choose Magnesium Fluoride,

$$\Gamma_{in} = \frac{\sqrt{n_a n_g} - n_c}{\sqrt{n_a n_g} + n_c} = -0.118$$

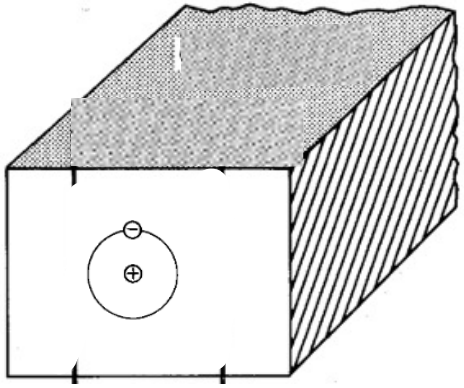
$$R = |\Gamma_{in}|^2 = 1.4\% \qquad d = \frac{\lambda_c}{4} = 99.6\ nm$$

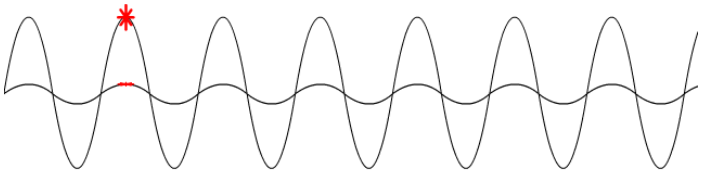
Note that without coating, (see figure)

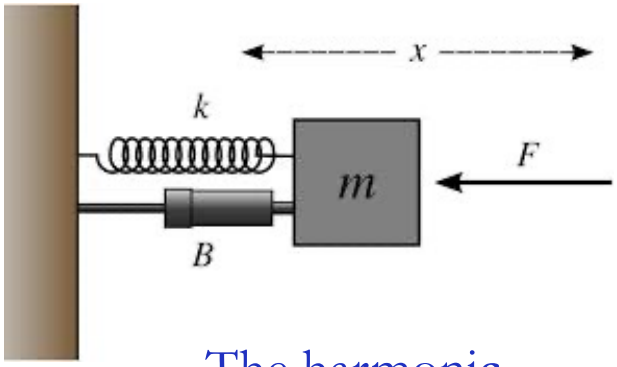
$$R = 4\%$$



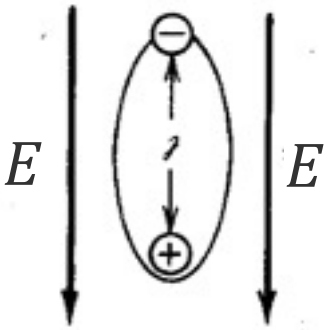
Light-matter Interaction



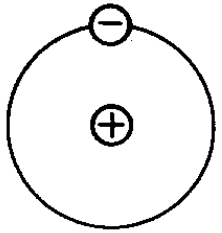




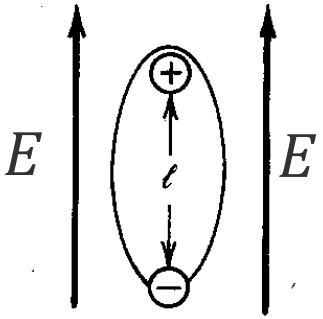
The harmonic oscillator



Atom polarized

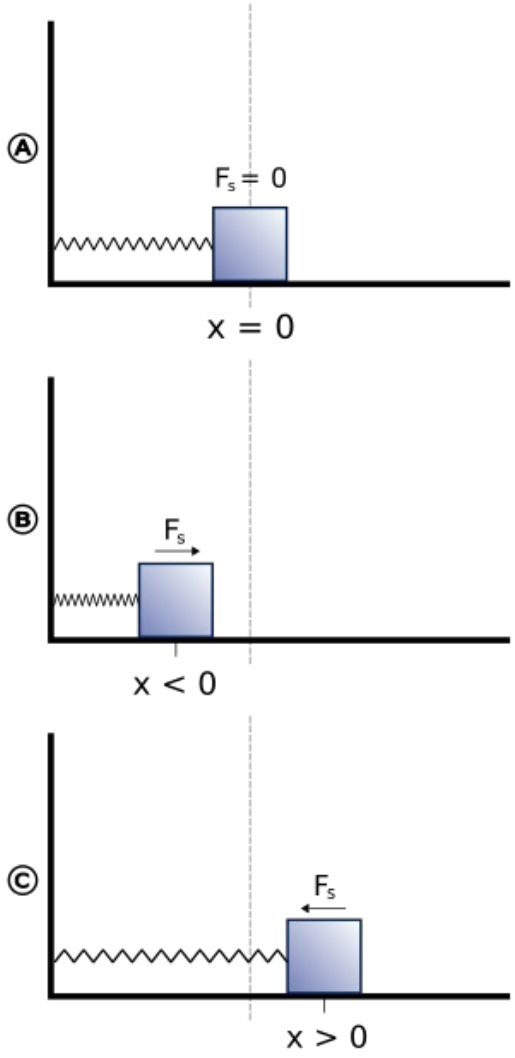


No Wave



Atom polarized

Light-matter Interaction



$$m\frac{\partial^2\vec{x}}{\partial t^2} + m\Gamma\frac{\partial\vec{x}}{\partial t} + m\omega_0^2\vec{x} = -q\vec{E}$$

- $m\frac{\partial^2\vec{x}}{\partial t^2}$: acceleration force
- $m \Rightarrow m_e$: mass of an electron
- $m\Gamma\frac{\partial\vec{x}}{\partial t}$: frictional force
- Γ: damping rate (loss/sec)
- $m\omega_0^2\vec{x}$: restoring force
- $\omega_0 = \sqrt{\frac{K}{m}}$: natural frequency
- $-q\vec{E}$: electric force

Fourier transform

$$m(-j\omega)^2\vec{x}(\omega) + m\Gamma(-j\omega)\vec{x}(\omega) + m\omega_0^2\vec{x}(\omega) = -q\vec{E}(\omega)$$

Simplify

$$\left(-m\omega^2 - j\omega m\Gamma + m\omega_0^2\right)\vec{x}(\omega) = -q\vec{E}(\omega)$$

Light-matter Interaction

$$\varepsilon_r(\omega) = 1 + \frac{\omega_p^2}{\omega_0^2 - \omega^2 - j\omega\Gamma} \qquad \omega_p^2 = \frac{Nq^2}{\varepsilon_0 m_e}$$

$$\varepsilon_r(\omega) = \varepsilon_r'(\omega) - j\varepsilon_r''(\omega)$$

$$\varepsilon_r'(\omega) = 1 + \omega_p^2 \frac{\omega_0^2 - \omega^2}{\left(\omega_0^2 - \omega^2\right)^2 + \omega^2\Gamma^2}$$

$$\varepsilon_r''(\omega) = \omega_p^2 \frac{\omega\Gamma}{\left(\omega_0^2 - \omega^2\right)^2 + \omega^2\Gamma^2}$$

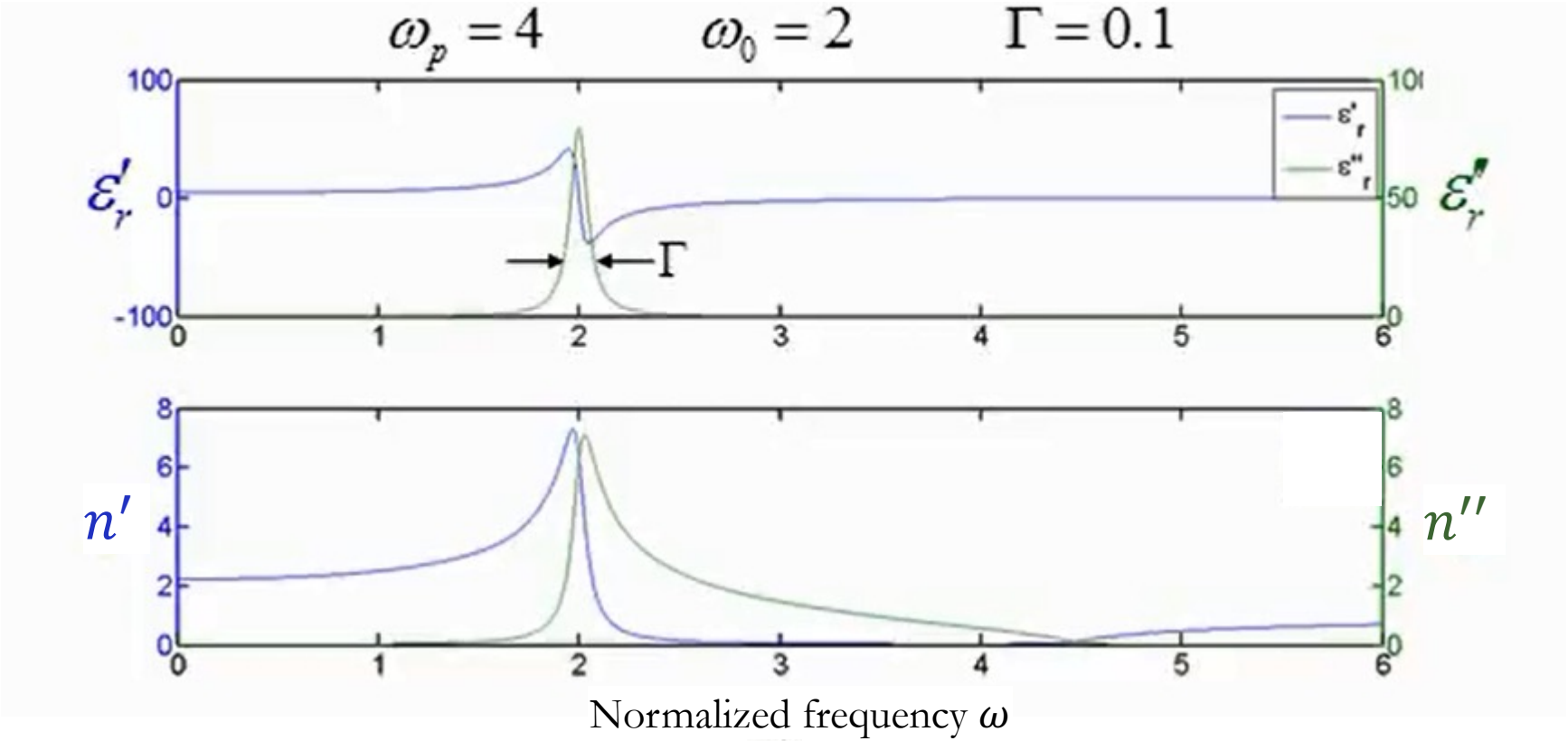
$$n = \sqrt{\mu_r \varepsilon_r} = n' - jn''$$ Complex Refractive index

$$(n')^2 = \frac{\varepsilon_r'}{2} + \frac{1}{2}\sqrt{(\varepsilon_r')^2 + (\varepsilon_r'')^2}$$

$$n'' = -\frac{\varepsilon_r''}{2n'}$$ Extinction coefficient

$$\alpha = 2\frac{\omega}{c}n''$$ Absorption Coefficient

Light-matter Interaction: Lorentz Model

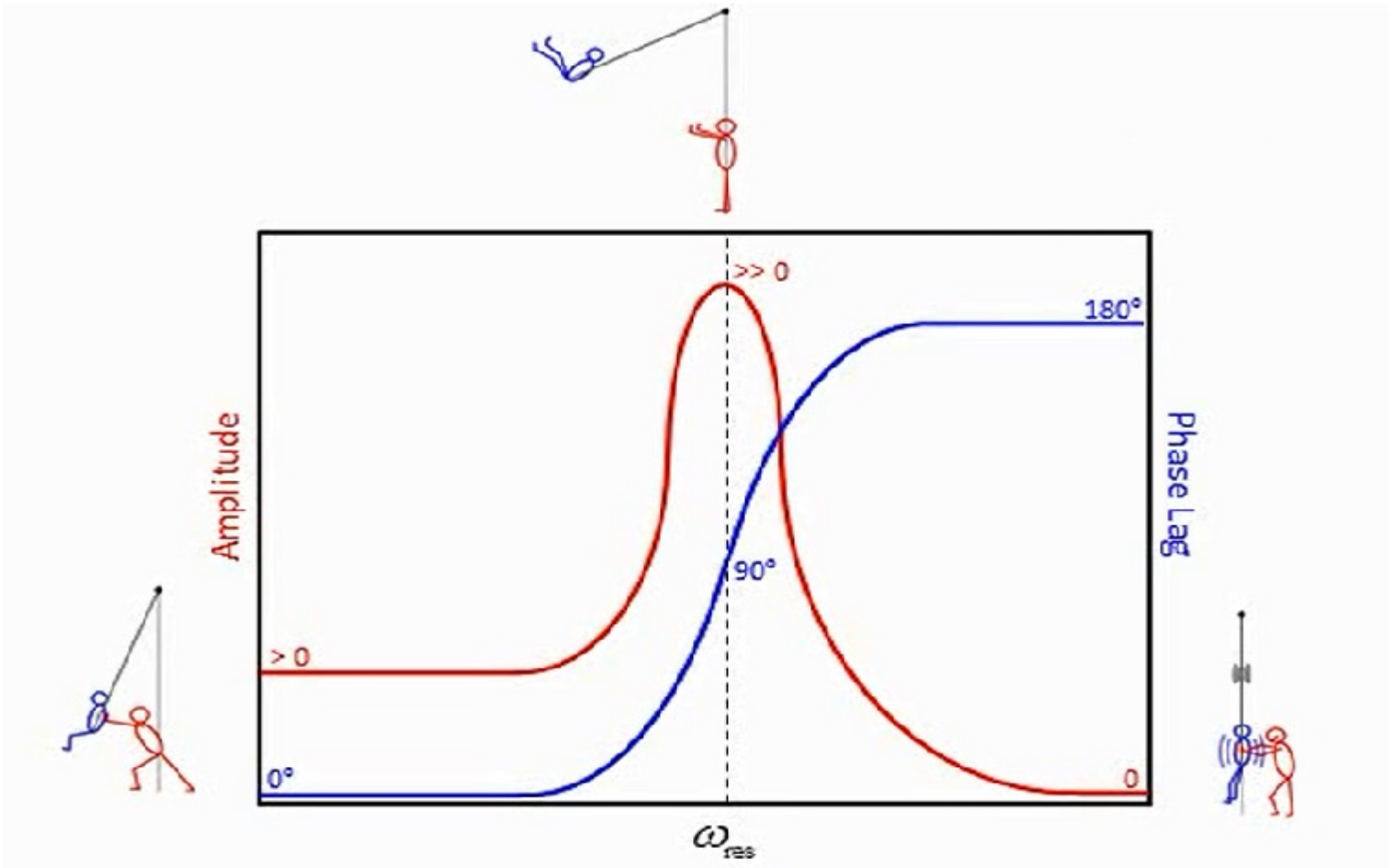


Loss is very high near resonance

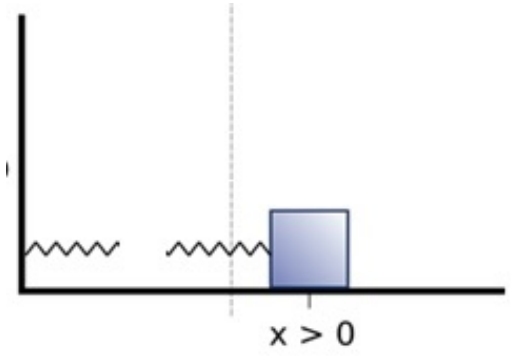
Damping rate determines width of resonance

Dielectric constant has a DC offset below resonance

Light-matter Interaction: Resonance



Light-matter Interaction: Metals



$$m\frac{\partial^2\vec{x}}{\partial t^2}+m\Gamma\frac{\partial\vec{x}}{\partial t}+m\omega_0^2\vec{x}=-q\vec{E}$$

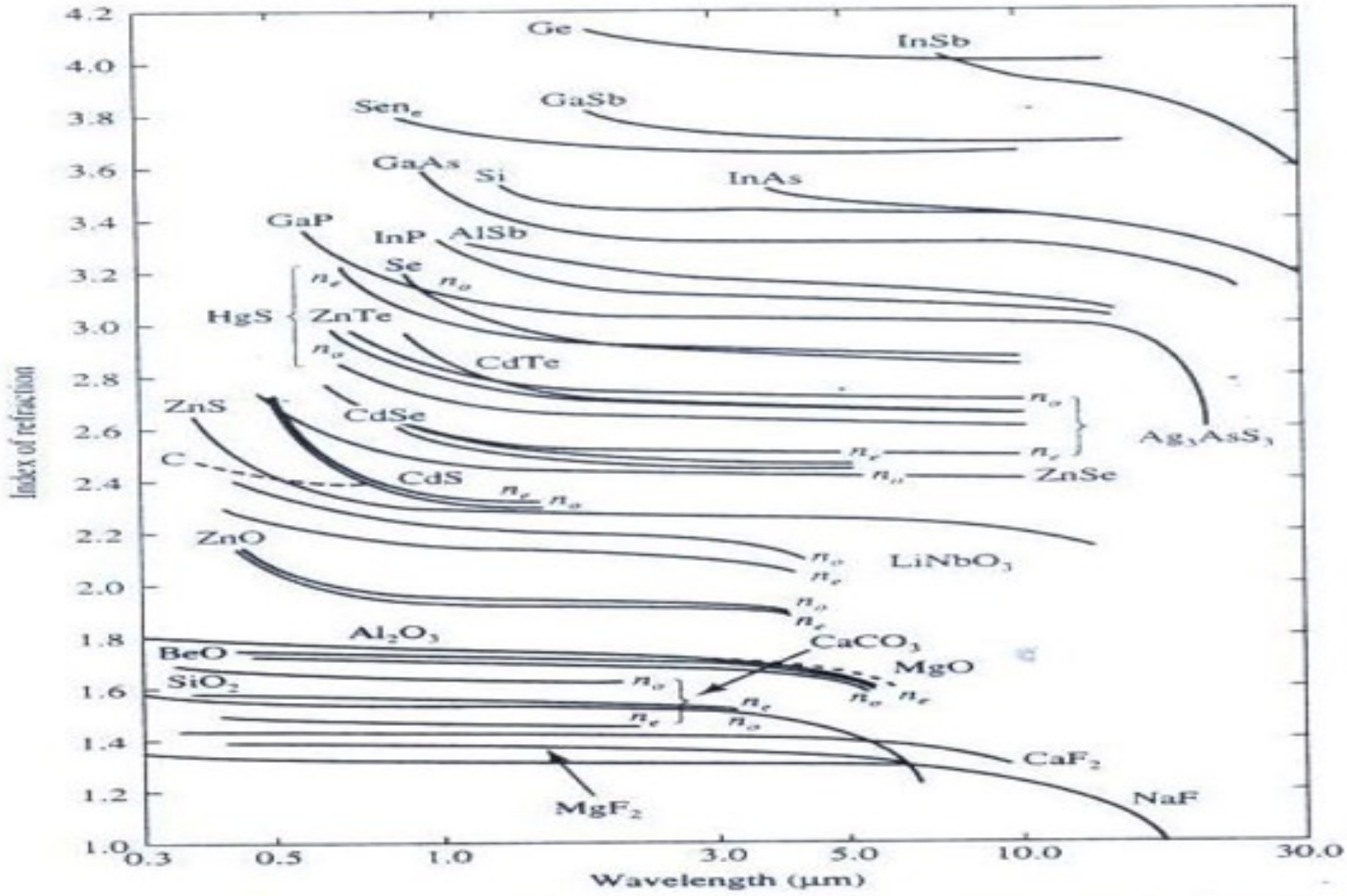
- $m\frac{\partial^2\vec{x}}{\partial t^2}$: acceleration force; $m \Rightarrow m_e$ mass of an electron
- $m\Gamma\frac{\partial\vec{x}}{\partial t}$: frictional force; Γ damping rate (loss/sec)
- $m\omega_0^2\vec{x}$: restoring force; $\omega_0=\sqrt{\frac{K}{m}}$ natural frequency
- $-q\vec{E}$: electric force

$$\varepsilon_r(\omega)=1-\frac{\omega_p^2}{\omega^2-j\omega\Gamma}$$

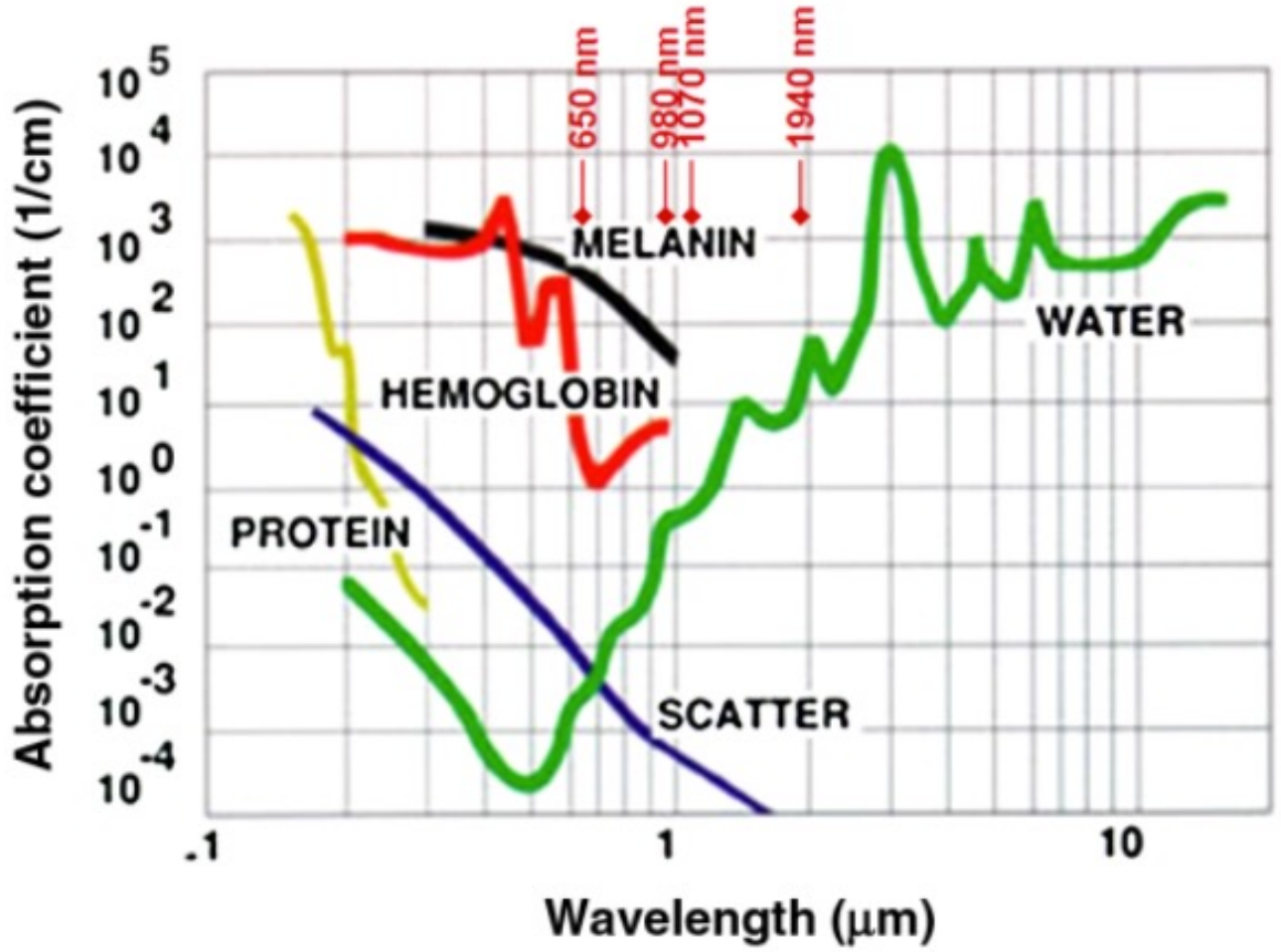
$$\varepsilon_r=\left(1-\frac{\omega_p^2\tau^2}{1+\omega^2\tau^2}\right)+j\left(\frac{\omega_p^2\tau/\omega}{1+\omega^2\tau^2}\right)$$

Drude Model

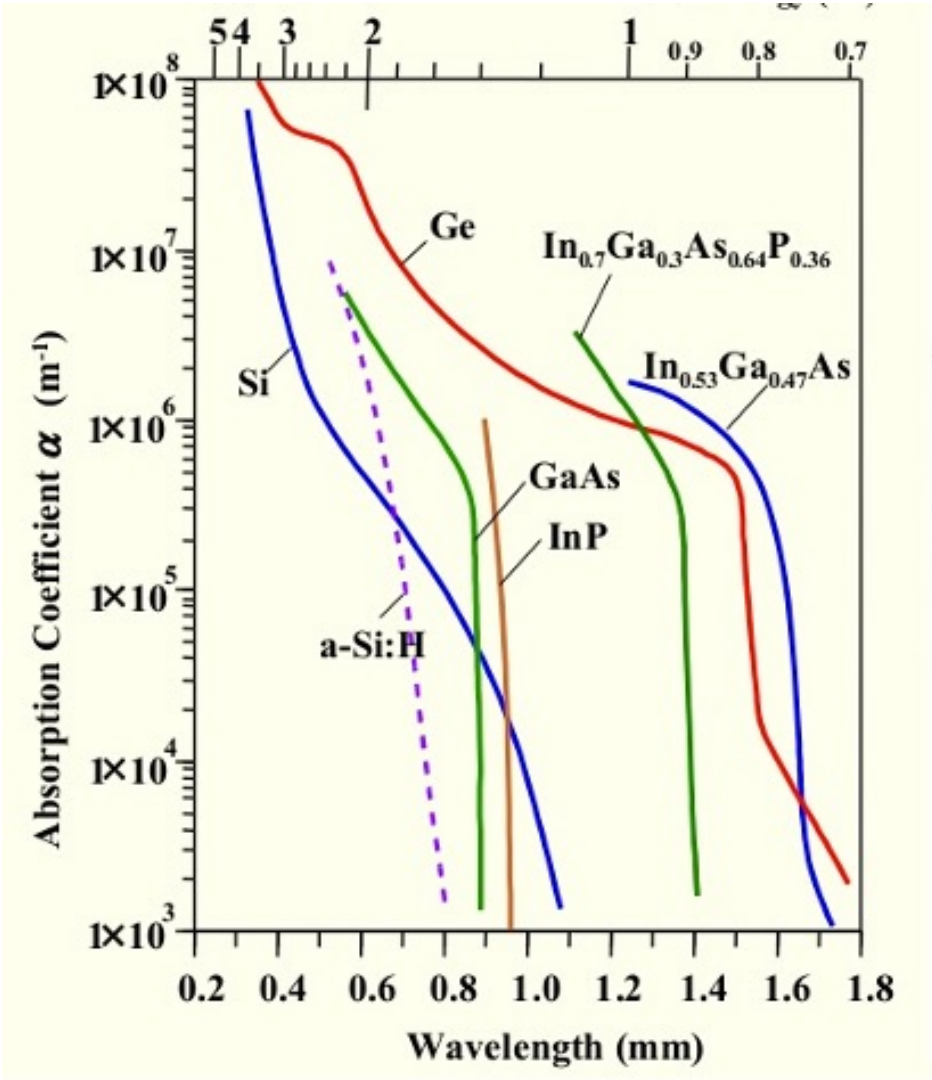
Light-matter Interaction: Examples
Refractive Index

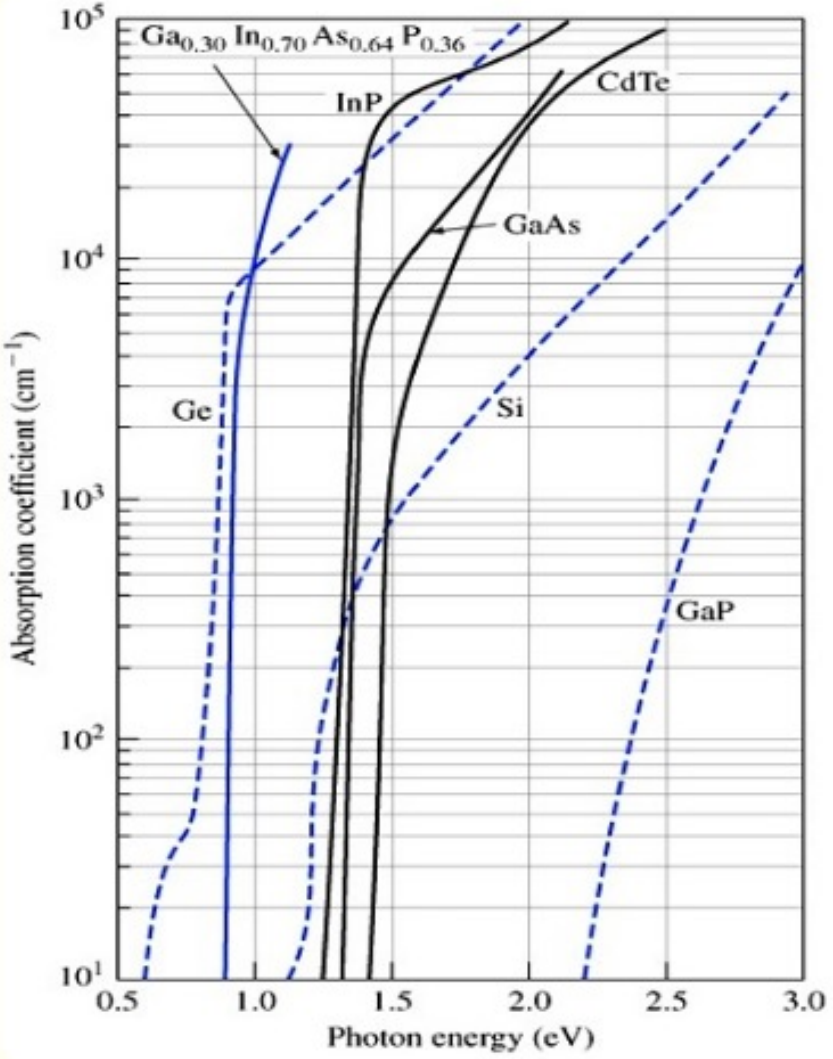


Light-matter Interaction: Examples
Absorption Coefficient



Light-matter Interaction: Examples
Absorption Coefficient





Q & A

Send me an email if you have any questions.